

Dataset



Air Pollution & Meteorological Data

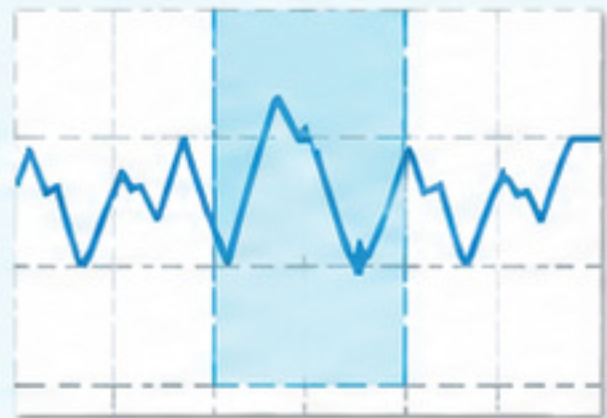
- Time
- CO (ppm)
- O₃ (ppb)
- NO₂ (ppb)
- SO₂ (ppb)
- PM₁₀ (μg/m³)
- PM_{2.5} (μg/m³)
- Temperature (°C)
- Dew Point (°C)
- Humidity (%)
- Wind Speed (km/h)
- Pressure (hPa)
- ...

Lag Features

	t	t-1	t-2	...	t-k
CO	x_t	x_{t-1}	x_{t-2}	...	x_{t-k}
O ₃	$x_t^{O_3}$	$x_{t-1}^{O_3}$	$x_{t-2}^{O_3}$	...	$x_{t-k}^{O_3}$
...	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
PM _{2.5}	$x_t^{PM_{2.5}}$	$x_{t-1}^{PM_{2.5}}$	$x_{t-2}^{PM_{2.5}}$	...	$x_{t-k}^{PM_{2.5}}$

Create lagged variables for target pollutant and other pollutants (e.g., 1, 2, 3, 7, 14, 30 days)

Rolling Statistics



Compute rolling statistics for multiple windows (e.g., 3, 7, 14 days)

- Rolling Mean
- Rolling Std
- Rolling Min
- Rolling Max

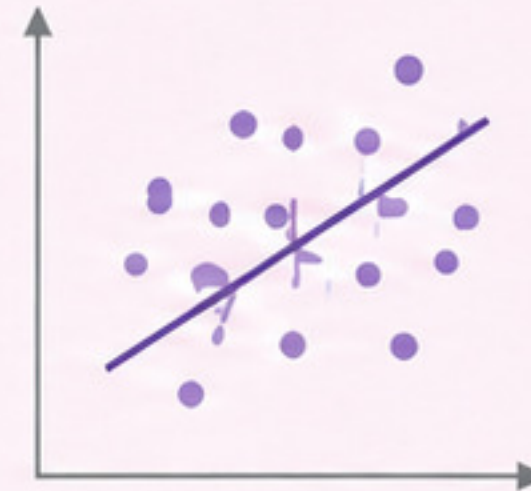
Cyclical Features



Extract time-based variables and apply sin-cos transformation

- Month (sin, cos)
- Day of Year (sin, cos)
- Day of Week (sin, cos)
- Week of Year (sin, cos)

Interaction Features

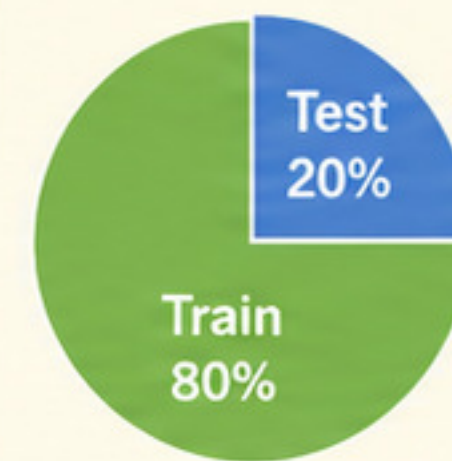


Create interaction terms between meteorological variables

e.g.,

- Temperature × Humidity
- Wind Speed × PM_{2.5}
- ...

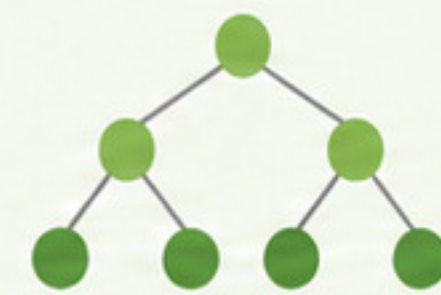
Train Test Split



Split data into training and testing sets (e.g., 80/20)

- Train set
- Test set

Random Forest & LightGBM



Random Forest



LightGBM

Train models using original + engineered features

Evaluation



Evaluate model performance

- R²
- RMSE
- MAE
- ...

Compare baseline vs. feature engineered models